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Innovative Approach for Assessing the Spatial Distribution and Environmental Impact of Fugitive Gases in Subsurface Aquifers at Oil Fields

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Abstract

Fugitive gases, also known as "stray gases," are naturally occurring gases released from oil refining and storage facilities. They can potentially contaminate groundwater aquifers near oil and gas fields. Despite their environmental significance, limited research has been conducted to investigate the extent of contamination caused by these gases. This study aims to investigate the presence, spatial distribution, and potential sources of fugitive gases in the groundwater of Kuwait's oil fields. Groundwater samples were collected from the northern, southeastern, and western regions of Kuwait's oil fields and analyzed for major and minor ions, trace elements, fugitive gases, and isotopes of methane (CH_4) and noble gases. Fugitive gas concentrations were highest in the northern Kuwait oil field, while hydrogen sulfide (H_2S) was more commonly detected in the southeast. Ethane (C_2H_6) was detected in a few samples, with the highest concentration observed in the northern region. Spatial analysis indicated that hydrocarbon gases migrate from oil fields through structurally weak zones into shallower aquifers. Isotopic analysis of methane ($\delta^{13}\text{C}$ and $\delta^2\text{H}$) confirmed a thermogenic origin with no microbial biogenic signatures, with $\delta^{13}\text{C}$ variations likely linked to the maturity of the source rock. The principal component analysis (PCA) identified a relationship between total organic carbon (TOC), pH, and C_2H_6 concentrations in groundwater. The study concludes that the distribution of fugitive gases is governed by pressure, temperature, conductivity, and subsurface structural features. Spatial distribution maps generated using GIS highlighted hotspots of higher gas concentrations in northern and western Kuwait, correlating with dissolved helium levels, while southeastern Kuwait showed lower concentrations. This study introduces a novel approach to assessing the occurrence and impact of fugitive gases in groundwater. It recommends further monitoring of additional wells in northern and western Kuwait, along with additional isotopic analysis and correlating these findings with subsurface structures.

Key Words: Investigation, Oil fields, Fugitive gas, Stable isotopes, Noble gases

Introduction

The rising demand of water for basic domestic needs has resulted in addressing the issues of contamination. Such assessments have become more critical in the regions with scarce freshwater resources. Although

studies have highlighted organic and inorganic contaminations in surface and groundwater resources, studies on contamination of fugitive gas in groundwater are scarce. "Fugitive gases" are also referred to as "Stray gas" in terminology adopted for naturally occurring gases liberated from refining or storing locations near the well-extracting oil or gas in an area where it is not expected (**Wisen et al., 2019**). Hydrocarbon gases have been reported to affect human health. Long-term exposure to such gases can affect the human lungs, eyes, and central nervous system (**Martínez et al., 2012**). Short-term exposure to an elevated concentration of these gases may lead to hypoxia and respiratory disorders (**Miller & Mazur, 1984**). Apart from these effects, exposure to hydrocarbon gases may also lead to rapid disorientation, dizziness, and narcosis (**Sugie et al., 2004**). The asphyxiant and explosive hazards posed by methane and related gases dissolved in residential well water pose a health risk when the gas migrates from the water into the air. There are two different methods of methane production: thermally produced, referred to as "thermogenic", and that of microbial origin, referred to as "biogenic".

- Thermogenic origin is generally produced by the pyrolysis of bigger molecules in hydrocarbon irrespective of the state of matter (solid, liquid, or gas) (**Seewald, 2023**).
- The microbial methane is generated by methanogen; which could result from two different processes:

- Hydrogenotrophic methane is produced by



- Reduction of CH_3 group from big organic molecules



Methane can also be generated by abiotic processes. Hydrogenation of CO_2 in the absence of biological components are mostly referred to as lithogenic (**Etiopé & Lollar, 2013**).

Studying noble gases in conjunction with fugitive gases helps in the determination of the source and origin of the temperature of the formation of the hydrocarbon gases (HCG). Noble gases are key indicators in the determination of HCG (**Wen et al, 2015**). Noble gases were used to focus on the determination of CH_4 sources (biogenic, atmospheric, or thermogenic and Lithogenic). Further, Leahy and Colwell (**Leahy & Colwell, 1990**) observed that the interaction between the oil and groundwater, mostly along their interface, affects the nature of accumulated HC. Subsequently, studies on this line (**Horstad et al., 1992**) have proved that the groundwater migration in the HC reservoir complex can enhance the microbes, thus leading to methane production and resulting in the biodegradation of HC fluid (**Stolper et al, 2014**).

Table 1—Studies on Migration of Gases from the Reservoirs Due to Different Factors

No	Inferred factor for migration	Country/ Region	Reference
1	Well Casing	Canada (Review)	<i>Bexte et al (2008)</i>
		USA	<i>Dusterhoft et al (2012)</i>
		New Mexico	<i>Chafin (1994).</i>
		Alberta, Canada	<i>Stempvoort et al (2005, 1996)</i>
		Ohio	<i>Bair et al. (2010)</i>
		Kansas	<i>Watney (2003)</i>
2	Abandoned wells	Ohio and Texas	<i>Kell (2011)</i>
		Fort Knox- Kentucky	<i>Lyverse and Unthank (1998)</i>
3	Salt Caverns	Kansas	<i>Allison (2001)</i>
4	Natural faults, weaker planes	Appalachian region	<i>Harrison (1985)</i>
		Pennsylvania	<i>Molofsky et al (2011)Osborn et al (2011)</i>

<i>No</i>	<i>Inferred factor for migration</i>	<i>Country/ Region</i>	<i>Reference</i>
		Clarendon-linden fault zones, New York	<i>Fountain and Jacobi 2000</i>
5	Micro-seepage		<i>Brown 2000Cartwright et al 2007</i>
6	Induced Fractures	Alberta, Canada	<i>Blyth 2008</i>
7	Induced Tremors		<i>Manga et al 2012</i>

To summarise, the literature review shows that previous studies have focused on the establishment of the baseline for all hydrocarbon exploration studies (Pik and Marty et al. 2009); development of control strategies for greenhouse gas emissions into the atmosphere (Vinson et al., 2017); and determination of enriched source rock regions (Miller et al., 2013). Studies on noble gases have helped in the identification of the isolated components of the hydrogeosphere, along with the extent of interconnection between the weaker zones and the fluid system (Lollar & Ballentine, 2009; Lippmann-Pipke et al., 2011). The importance of the study stems from the fact that literature shows that water wells in oil fields tend to release a continuous stream of hydrocarbons from deep source rock horizons towards shallower zones (Tolstikhin et al., 2017). Accordingly, the isotopes of radiogenic noble gases tend to concentrate in interstitial water, moving towards the hydrocarbon source region. Groundwater may act as a receptor or a pathway for fugitive gases that could originate from surface sources, pass through the aquifer's unsaturated zone, dissolve in water, and eventually reach the water table. They may also travel through groundwater from biogenic subsurface reservoirs or radiogenic sources (Lau et al., 2016). Furthermore, these gases may migrate laterally or vertically to building basements as gases dissolved in groundwater (Morgan et al., 2014). Despite their environmental significance, limited research has been conducted to investigate the extent of contamination caused by these gases. This study aims to investigate the presence, spatial distribution, and potential sources of fugitive gases in the groundwater of Kuwait's oil fields. The study is the first of its kind in the region involving the development of a methodology to address the issue of fugitive gas contamination in the Kuwait Group and the Dammam Formation aquifers, targeting three specific sites located in the oil fields of North Kuwait, Southeast Kuwait, and West Kuwait.

Study area

Located in a dry area, Kuwait experiences cool winters and hot, dry summers. The country receives an average of 110-120 mm of precipitation per year, most of it falls between May and October (Schulz et al., 2015). The annual evaporation rates exceed the annual rainfall, varying from 2500 mm in the coastal areas to 4500 mm inland. The ground elevation in Kuwait ranges from 300 m above mean sea level (AMSL) in the SW to sea level along areas that border the Arabian Gulf, except for the Jal Al-Zor Escarpment, which borders Kuwait Bay and rises more than 127 m AMSL. Hydrologically, Kuwait has two major aquifers: the Dammam Formation (DF) Aquifer within the Hassa Group (HG) and the unconformably overlying Kuwait Group (KG) Aquifer (Fig. 1). From the bottom, the DF underlies the whole of the State of Kuwait and forms the main aquifer in Kuwait. The DF mainly comprises carbonates with evaporites, shale deposits, and minor chert (Misak & Hussain, 2022). The DF exists with heterogeneous lithology and saline due to a gradual decrease in resistivity from southwest to northeast. Kuwait is predominately covered by the KG, which is composed of silts, clay, sandstone, gravel and sand (Al-Fahad et al., 2002) and was reported to be deposited during the Early Oligocene to the Pleistocene, according to Amer and Al-Hajeri (2019).

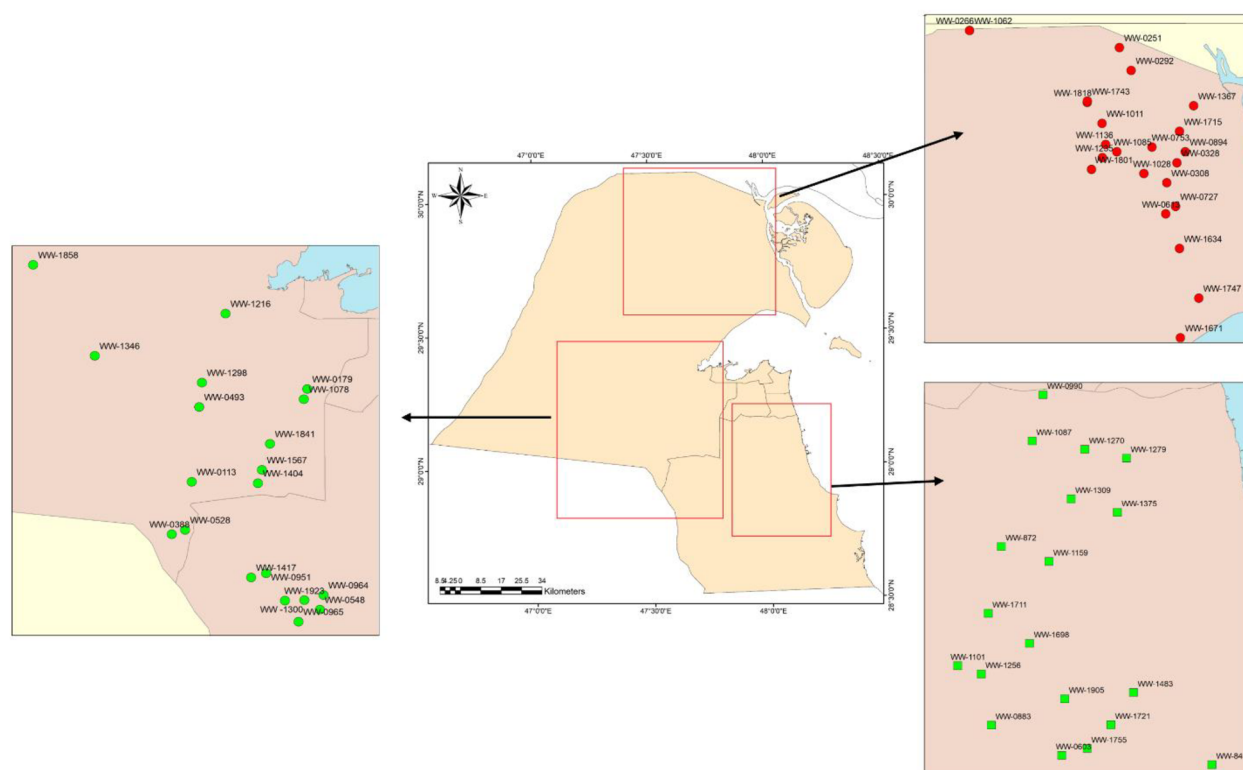


Figure 1—The sampling locations considered for the study represent different locations adjoining the oil fields of north, west and south-east Kuwait.

The geology of the study area in northern Kuwait (Fig. 1) is covered mainly by the upper Dibdibba Formation. The lithology of the northern part of Kuwait is predominantly characterized by siliciclastic sediments and sedimentary rock units, with the presence of the Dibdibba Formation and the Eocene limestone and chert of the Dammam Formation (AlRefaei et al., 2022). The northern part of Kuwait is home to significant oil fields, generally known as the Sabriyah and Raudhatain oil fields, which include the North Kuwait Conventional Oil Field and the North Kuwait Heavy Oil Field (Al-Saad et al., 2013). The geology of the region has played a crucial role in the formation of these oil fields, with the presence of rich source rocks, high-quality reservoir rocks, and efficient seal rocks contributing to the development of multiple petroleum systems in Kuwait (Naqi et al., 2022).

The lithology in the southern part of Kuwait is characterized by a mix of sediments, including carbonate deposits, sandstones, siltstones, and shales, with the presence of the Eocene Dammam formation is of the Jurassic period. The Greater Burgan field, which includes the Burgan, Magwa, and Ahmadi fields, is located in the southeast region of Kuwait and is the biggest and oldest oil field in the country. The fault planes in the Burgan Oil Field is observed in a NW-SE orientation. Gases will be able to migrate to shallow aquifers more easily if there are weaker structures in the area surrounding large oil fields (Sabarathinam et al., 2020). The transport of gases from deeper reservoir regions to shallow groundwater is facilitated by weaker zones (Mukhopadhyay et al., 2007). Siliciclastic sediments, with some carbonate deposits and a mix of other sedimentary rock units, characterize the western part of Kuwait. The Dibdibba Formation and Jurassic sediments are significant components of the lithology in this region. Minagish and Umm Gudair are the two most significant fields in West Kuwait, with other smaller fields scattered throughout (Davies et al., 2000). The Minagish Formation is a part of the Umm-Guddair field, including upper, middle, and lower sections. The upper Minagish section is a non-reservoir with irregular porosity developments. In contrast, the lower Minagish section is devoid of hydrocarbons yet has good reservoir rock characteristics, and the middle Minagish is a good hydrocarbon-bearing reservoir. The Abdaliyah region was observed to have dolomitic

limestone formation (Fig. 2). From top to bottom, the unsaturated zone in the KGF is characterized by a high resistivity, a low resistivity zone in the saturated KGF aquifer, increasing resistivity in the Dammam and Rus formations due to their lower porosity and water content, and decreasing conductivity was observed once the Umm Er Radhuma (UER) reached its higher porosity and salinity. There are three distinct levels within the Abdaliyah area's Dammam Formation. The upper level consists of massive chalky dolomite, the middle level of laminated dolomite, and the lowest level of nummulitic limestone. Sediments of these three members have been developed in three different sediments geological paleo-environments.

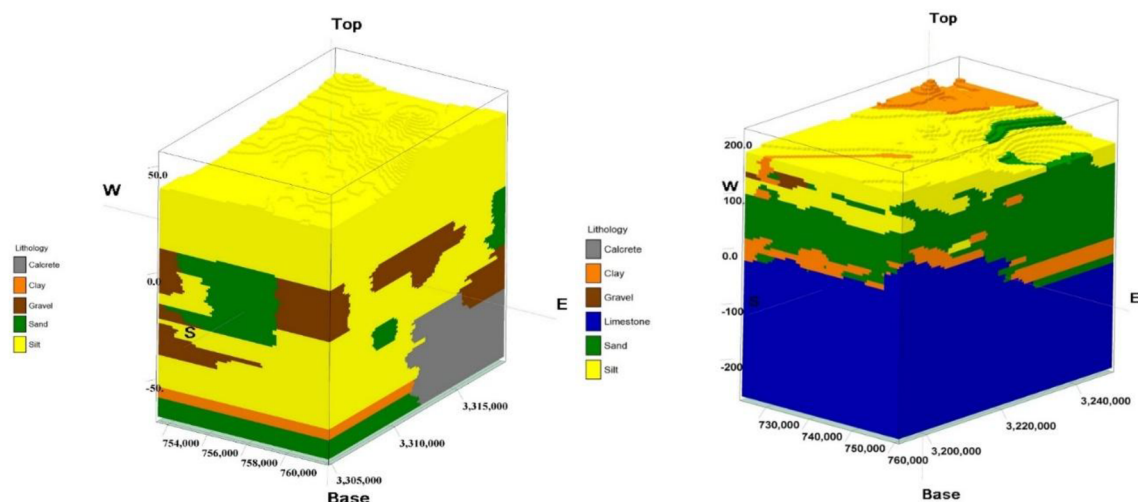


Figure 2—Three-dimensional section of the north and west Kuwait well field region, constructed with the litholog data obtained for the selected wells (The lithology was generalized based on the dominant types across the section).

Methodology

Three different oil field sites were selected for one-time sampling of groundwater from different aquifers (KGA and DM): They represented the North Kuwait oil field (NK), Southeast Kuwait (SEK) and West Kuwait (WK). Samples from boreholes tapping the DM (in WK and SEK) and from the Kuwait Group (NK) were collected (Fig. 1). Based on the availability of sampling wells 22 samples were collected for NK, 20 each in WK and SEK. The purging continued until three volumes of the well water and stabilization of pH and Electrical Conductivity (EC) of groundwater were completed. After purging of a well, groundwater samples were collected in bottles and copper tubes. Samples to be used for analyses of major cations (Ca, Mg, Na, K), major anions (SO_4 , Cl, CO_3 , HCO_3 , NO_3) and total organic carbon (TOC) were collected in appropriate polyethylene and glass bottles, whereas, the samples to be used for analyses of noble gases were collected in copper tubes. The collected groundwater samples were analyzed for major cations (Ca, Mg, Na, K); major anions (SO_4 , Cl, CO_3 , HCO_3 , NO_3); TOC adopting standard methods of analysis (APHA, 2012); noble gas (He, Ne, Xe, Ar, and Kr); ^2H ; and ^{18}O . The major cations, anions, and TOC analyses were conducted in the KISR, Kuwait WRC laboratory. Noble gases He, Ne, Ar, Kr and Xe in Fluid Extraction Noble Gas Separation Quadruple Mass Spectrometer at the University of Nebraska Lincoln Laboratory. Samples for fugitive gases CH_4 and C_2H_6 and isotopes and at International Laboratory, National Environmental Isotope Facility, UK.

Results

The average pH of the samples in North Kuwait tends to be alkaline though few samples reflected with acidic pH. The results of EC indicate the brackish nature of the groundwater as few samples reflected EC above average value of $11605\mu\text{S}/\text{cm}$ (Table 2). The order of dominance of the major ions

are $\text{Cl}^- > \text{Na}^+ > \text{Ca}^{2+} > \text{SO}_4^{2-} > \text{HCO}_3^- > \text{NO}_3^- > \text{K}^+ > \text{Br}^-$, respectively, signifying that Cl^- is the major ion followed by Na^+ in the groundwater. The results of EC reflect that a few wells indicate fresh water while few are brackish in nature, as reflected by EC's average value of $10718 \mu\text{S}/\text{cm}$ in the Southeast oil field. The order of dominance of the major ions are $\text{Cl}^- > \text{Na}^+ > \text{SO}_4^{2-} > \text{Ca}^{2+} > \text{HCO}_3^- > \text{CO}_3^{2-} > \text{NO}_3^- > \text{K}^+ > \text{Br}^-$, respectively, signifying that Cl^- is the major ion followed by Na^+ in the groundwater. Similarly, the west Kuwait samples had pH ranging from acidic to alkaline with an average EC of $8222 \mu\text{S}/\text{cm}$. The order of dominance shows that $\text{Cl} > \text{SO}_4 > \text{Na} > \text{Ca} > \text{Mg} > \text{HCO}_3 > \text{NO}_3 > \text{K} > \text{Br} > \text{CO}_3$.

Table 2—Minimum, Maximum, and Average Values of Physical and Chemical Analytes of the Collected Samples

UNITS	North Kuwait			South East Kuwait			West Kuwait		
	Min	Max	Avg	Min	Max	Avg	Min	Max	Avg
pH	6.99	9.89	8.11	7.23	10.03	9.05	6.302	8.328	7.49
EC $\mu\text{S}/\text{cm}$	4590	53800	11605	302	16070	10718	3244	38900	8222
TDS mg/l	3168	37500	8051	193	10300	6859	2080	24900	5262
TOC mg/l	0.51	2.24	1.32	0.82	6.59	1.79	0.22	9.4	1.89
Silica mg/l	0.25	22.5	9.64	0.02	32.8	6.76	0	24.9	15.97
S^{2-} mg/l	<0.5	2	2	1.2	16	8.43	<0.01	13.2	13.20
Li^+ mg/l	0	1	0	0.01	0.44	0.23	0	0.39	0.12
Na^+ mg/l	924	9512	2132	16	2872	1550	414.5	3426	893
NH_4^+ mg/l	0.48	10.7	5	1.92	84	17	0	5.5	0.64
K^+ mg/l	7.11	132	20	0.99	102	35	8.82	80.2	20.5
Ca^{2+} mg/l	13.3	2694	491	16.9	1006	599	58.9	3203	554
Mg^{2+} mg/l	5.61	1051	155	4.89	356	192	20.3	829.8	190
F^- mg/l	0.79	3.2	1.8	0.31	1.49	0.83	0.18	1.93	1.38
Cl^- mg/l	1323	22314	3511	76.1	5234	3449	436	13623	2142
NO_2^- mg/l	<0.01	5.3	5.3	<0.01	1	1	0.153	1.85	0.91
Br^- mg/l	1.3	108	8.8	0.3	20.1	9.65	0.99	73.2	8.02
NO_3^- mg/l	0.9	148	52.4	0.25	78.2	14.1	0	369.4	54.48
PO_4^{3-} mg/l	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	0	27.43	6.77
SO_4^{2-} mg/l	391	2553	1524	3.94	1809	999	104	1740	1248
CO_3^{2-} mg/l	3	46	18.8	2	78	28.8	0	30	5.45
HCO_3^- mg/l	13.9	192	69.8	0	232	38	0	200	83.15
Al $\mu\text{g}/\text{l}$	12	25.9	19.7	6.35	39.6	21.5	17.2	144	33.84
Ba $\mu\text{g}/\text{l}$	6.58	55.9	20.7	7.31	10718	536	9.67	56.54	20.29
Co $\mu\text{g}/\text{l}$	1.28	2.55	1.93	1.62	2.39	1.95	184	5212	1139
B $\mu\text{g}/\text{l}$	635	4023	1902	426	6859	1355	0.01	4.22	1.55
Fe $\mu\text{g}/\text{l}$	1.11	355	56.4	1.34	169	25	1.22	229	48.11
Pb $\mu\text{g}/\text{l}$	0.07	1.06	0.54	0.19	8.43	2.39	<0.002	0.85	0.85
Sr $\mu\text{g}/\text{l}$	3672	79954	17685	806	41348	22023	3982	82036	22568
Th $\mu\text{g}/\text{l}$	0.55	7.57	5.1	0.41	1550	78	3.63	7.65	5.98
V $\mu\text{g}/\text{l}$	0.49	16.6	3.97	0.5	17.4	2.18	0.05	7.54	2.68
CH_4 $\mu\text{g}/\text{l}$	0	688	6.81	0.25	101.8	15.57	0.27	419	29.55
C_2H_6 $\mu\text{g}/\text{l}$	0	386	2	0	7.17	0.89	0	3.49	0.44

Though samples from all the well fields had CH₄ representations, North Kuwait had more samples with methane (Fig. 3), and relatively fewer samples were detected with methane in SE and West Kuwait. The highest concentration was observed in north Kuwait wells WW0308B (688 mg/L) and WW0328B (322 mg/L); the west Kuwait well WW0179 also had a higher concentration of methane (419mg/L). It should also be noted that salinity plays a major role in the dissolution of fugitive gases in the groundwater apart from temperature (Avrahamov et al., 2014).

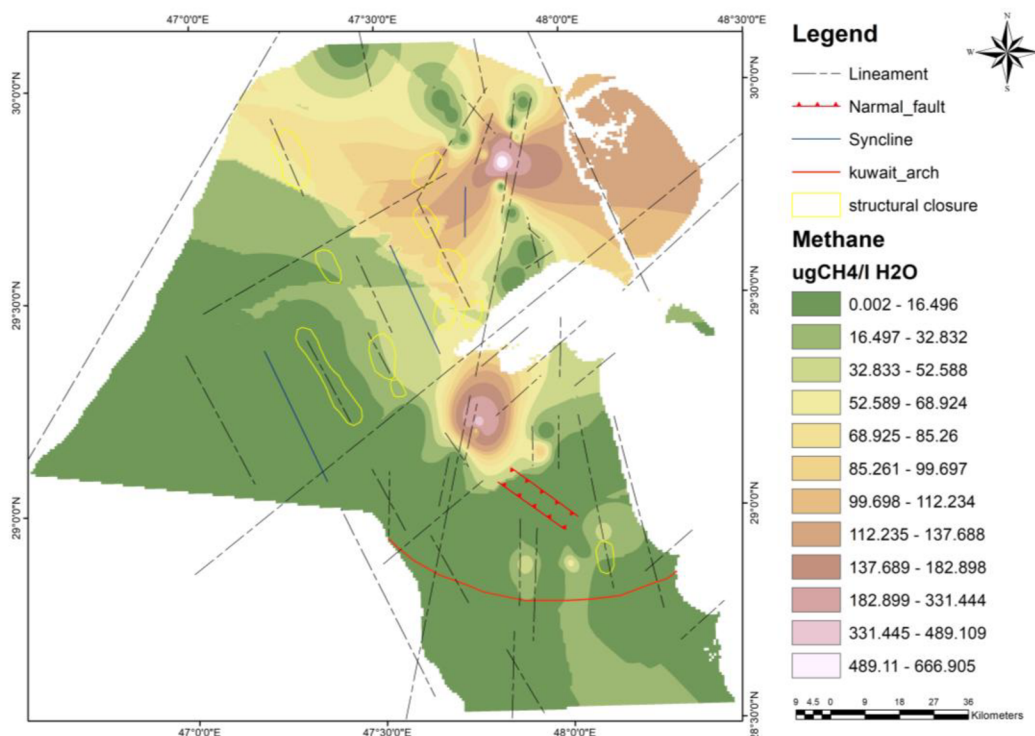


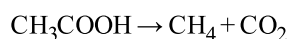
Figure 3—Spatial distribution of methane based on the samples collected from three well fields plotted against surficial geological structures.

Methane:

The geochemical water types derived based on the composition of water samples indicating the predominance of cations and anions in water show that the Na-Cl-SO₄ water type is dominant in north Kuwait, Na-Ca-Cl in southeast Kuwait, and Na-Ca-Mg-Cl-SO₄ type in west Kuwait. Thus, it is inferred that the dominant water type of the three well fields is different. The higher CH₄ in the groundwater was observed in the Na-Cl-SO₄ type in north Kuwait, Na-Ca-Cl in southeast Kuwait, and Na-Cl type in West Kuwait.

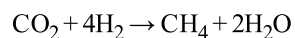
Nature fractionates materials isotopically (Criss, 1999), and Chemical reactions favour the lighter isotope. Methane has a wide range of $\delta^{13}\text{C}$ (100‰–20‰) and $\delta^2\text{H}$ (400‰–100‰), resulting from kinetic isotope fractionation from microbe-generated natural gas and thermocatalytic cracking of organic molecules (Schoell, 1983). Three major types are Microbial, thermogenic, and abiogenic natural gas (Whiticar, 1994). Methane, produced by fermentation of methyl-type substrates such as acetate or CO₂ reduction in anoxic settings, dominates microbial gas (Whiticar, 1999)

Methane is produced naturally in near-surface, mainly watery habitats (e.g., swamps, sewers, compost, landfills) by the generalized reaction in Equation:



3

The following equation (4) shows CO₂ reduction in anoxic conditions:



4

Organic matter (kerogen) undergoes thermocatalytic bond breakage at rising temperatures (i.e. burial depth), releasing the thermogenic gas. According to **Schoell (1983)** and **Whiticar (1996)**, the type of source rock determines the final hydrocarbon makeup and isotope ratios. Combining compositional and compound-specific isotope data makes it possible to differentiate between distinct source rocks (marine, terrestrial, or lacustrine) and their maturity (the amount of alteration; **Tilley & Muehlenbachs, 2012**). Even at low temperatures, the carbon isotopes of thermogenic gases, such as methane, ethane, and propane, often fall within the range of 50–25 per mill, 40–20 per mill, and 40–20 per mill, respectively (**Rowe & Muehlenbachs, 1999**). The $\delta^{13}\text{C}$ in microbial methanogenesis has a definite signature; microbial methane depletes significantly in $\delta^{13}\text{C}$, often $<60\%$ (**Faber & Stahl, 1984**), and Microbial ethane is typically $<40\%$ in $\delta^{13}\text{C}$ (**Rosenfeld & Silverman, 1959**). The $\delta^{13}\text{C}$ of thermogenic methane typically varies from -55% to -20% , distinguishing it from microbial gas (**Fig. 4**). The normal distribution of isotope readings for any maturity level is $\delta^{13}\text{C}_1 < \delta^{13}\text{C}_2 < \dots < \delta^{13}\text{C}_{\text{kerogen}}$. The $\delta^{13}\text{C}$ of all C1–C5 components increases with maturation, approaching the source kerogen until only C1 remains. Kerogen $\delta^{13}\text{C}$ values vary from -35% to -20% depending on the source and geological age (**Lewan, 1986**).

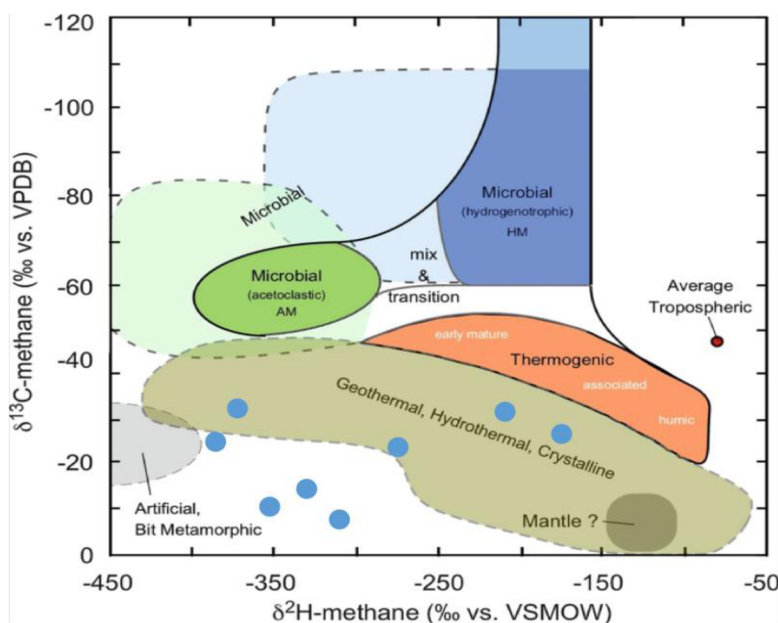


Figure 4—Sources of Fugitive gases in groundwater samples from north Kuwait.

A plot was developed by **Whiticar (1986, 1999)** to determine the methane origin with the aid of $\delta^2\text{H}$ and $\delta^{13}\text{C}$ values of Methane. A transitional region or a common boundary for thermogenic and biogenic processes can range from -45 to -60 per mil of $\delta^{13}\text{C}$, depending on $\delta^2\text{H}$ values. In the $\delta^2\text{H}$ region, it varies from -152 to -275 per mil. The samples of north Kuwait indicate that they are mostly of thermogenic origin (**Fig. 4**), similar to the Kerogen, but it has to be further correlated with the age of the formation.

Noble Gases.

^3He is mainly contributed from the Atmosphere, referred to as the Tritogenic component as it is derived from the decay of Atmospheric tritium. ^4He indicates the Radiogenic component as this isotope of Helium is mainly derived from the decay of minerals in the rocks - lithogenic sources. The noble gas analysis in the groundwater samples indicates that He had higher concentrations relatively than $\text{Ne} > \text{Kr} > \text{Xe}$. The Helium concentration was higher in samples in the northeastern and southern parts of the country. A definite range of $^3\text{He}/^4\text{He}$ and Ne/He ratios helps to determine the signatures of atmospheric, crustal, and mantle

sources. Significant variations in the He isotope ratios are observed between the mantle and the crustal Helium (**Mamyrin & Tolstikhin, 1984**). The Ratio of Helium Isotopes (R_a) in different end members (Atmosphere, Crustal and Mantle) and R_s , represents the ratio of respective sample R_s to atmospheric end member. R_{air} indicates the ratio of the Helium ($^3\text{He}/^4\text{He}$) in air, and R_s indicates the ratio in samples. Comparing the ratio of R_s/R_{air} , it is easy to fingerprint the atmospheric, crustal or mantle component of Helium in the samples (**Rashid et al., 2023**). These definite ratios relative to atmospheric He is 1.384×10^{-6} ($1R_{air}$), 1.5×10^{-8} ($0.02R_{air}$), and 1.15×10^{-5} ($8R_{air}$), for air, crust, and mantle sources, respectively (**Castro, 2004**). Air-saturated water (ASW) at an equilibrium solubility with the atmosphere would have a ^4He concentration of $4.475 \times 10^{-8} \text{ cm}^3 \text{ STPg}^{-1} \text{ H}_2\text{O}$ at 19°C (**Cey et al 2009**). The ratios of $^3\text{He}/^4\text{He}$ (R_A) and sample helium ratio to atmospheric ratio also serve as valuable indicators for determining the geological processes and groundwater dynamics (**Kulongoski et al 2005**). If the R_s/R_{air} is around 1, it indicates the dominant atmospheric component, as observed in samples WW1801, WW0292, WW0266 and WW1062. ^{20}Ne , is abundant in the air, and the occurrence of ^{20}Ne in the crust and mantle is negligible (**Rashid et al., 2023**). Hence, the Ne/He ratio would help to determine the Helium sources in the samples (Fig. 5). Thus, plotting the samples with these values indicates that samples WW1801, WW0292, WW0266 and WW1062 have atmospheric signatures, and the other samples have more crustal or lithogenic influences as inferred from the isotopes of C & H in the groundwater samples

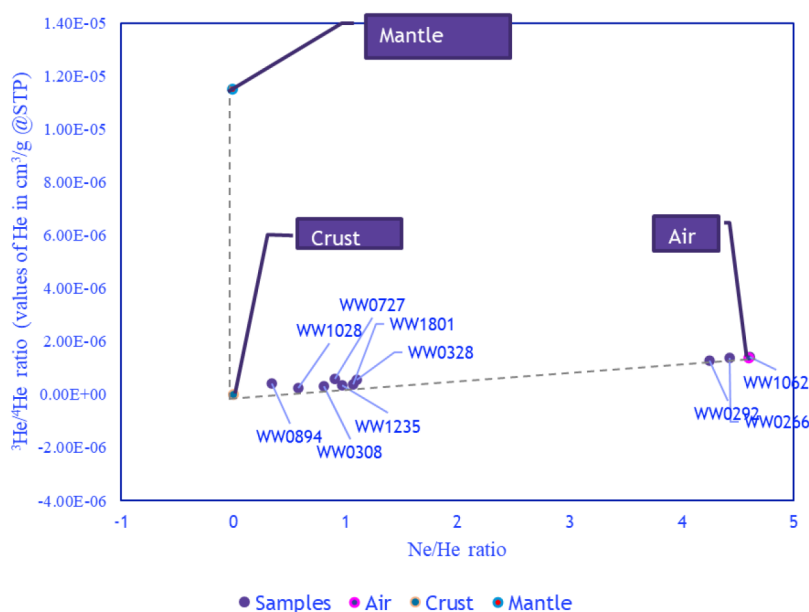


Figure 5—Determination of the fugitive gas sources through Noble gas isotope ratios of Helium and Ne/He.

Geochemical Process

Several processes in the groundwater with respect to the research locations are reflected in the principal component analysis (PCA) of the geochemical results (Fig. 6). The analytes and their methods constitute factor, and the broad lines reflect each research field. Factor 1's analyte contribution is consistent across all regions of study, as it depicts the ions dissolving from the aquifer matrix and resulting in brackish groundwater (**Sabarathinam et al., 2025**). In contrast, factor 2's analyte contribution is silicate weathering in North Kuwait and hydrogenation in Southeast and West Kuwait. In the north of Kuwait, factor 3 represents the dissolution of fluoride carbonate; in the southeast, it represents the dissolution of sulphate minerals; and in the west, it represents the dissolution of silica minerals, which are unfavored by ethane. Ethane could not be produced in North Kuwait's factor 4 due to the organic carbon. Anaerobic oxidation, on the other hand, is seen in the southeast, and in west Kuwait, the process is extremely complicated, including many different

processes. Methane oxidation was detected in North Kuwait's factor 5. On the other hand, in the Southeast, methane was linked to Ca^{2+} and F^- (Sabarathinam et al., 2025).

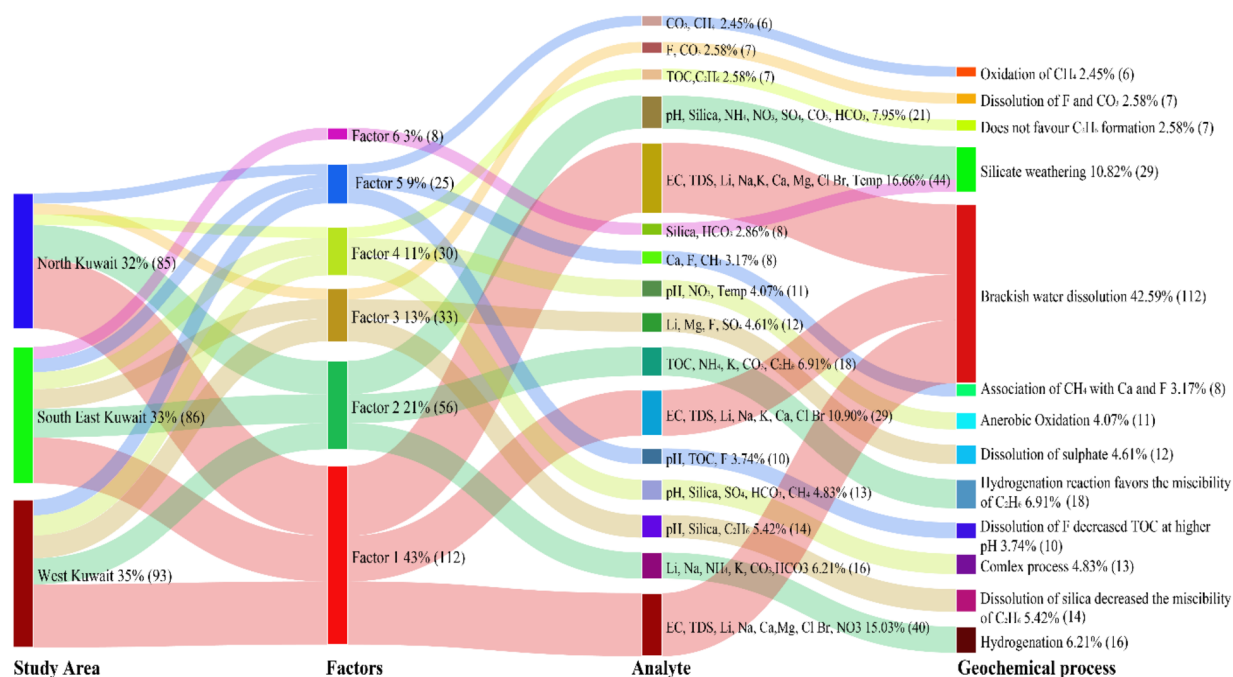


Figure 6—Summary of the major processes governing the Geochemistry of the three well fields and their relation to CH_4 and C_2H_6 concentration (Linkage indicated in Blue and Dark Blue colours). The Sankey chart signifies the linkage of study data at different locations, factors, different types of analysts represented in groundwater and the sources for each factor (after Sabarathinam et al., 2025).

Conclusions

According to the results of the spatial analysis, hydrocarbon gases move from the oil fields to the shallower aquifers through structural features. There were no signs of microbial biogenic processes in the isotope study of methane ($\delta^{13}\text{C}$ and $\delta^2\text{H}$), indicating that it originated thermally. The fluctuations in $\delta^{13}\text{C}$ are most likely related to the maturity of the source. Groundwater C_2H_6 concentrations, pH, and total organic carbon (TOC) were found to be related by principal component analysis. Pressure, temperature, conductivity, and subsurface structural characteristics control the spread of fugitive gases. Northern and western Kuwait displayed hotspots of higher gas concentrations, coinciding with dissolved helium levels. Southeastern Kuwait, on the other hand, showed lower concentrations. Additional wells in northern and western Kuwait should be monitored, isotope analysis should be conducted, and the results should be correlated with subsurface structures. This work suggests a new way to measure the presence and effects of fugitive gases in groundwater.

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